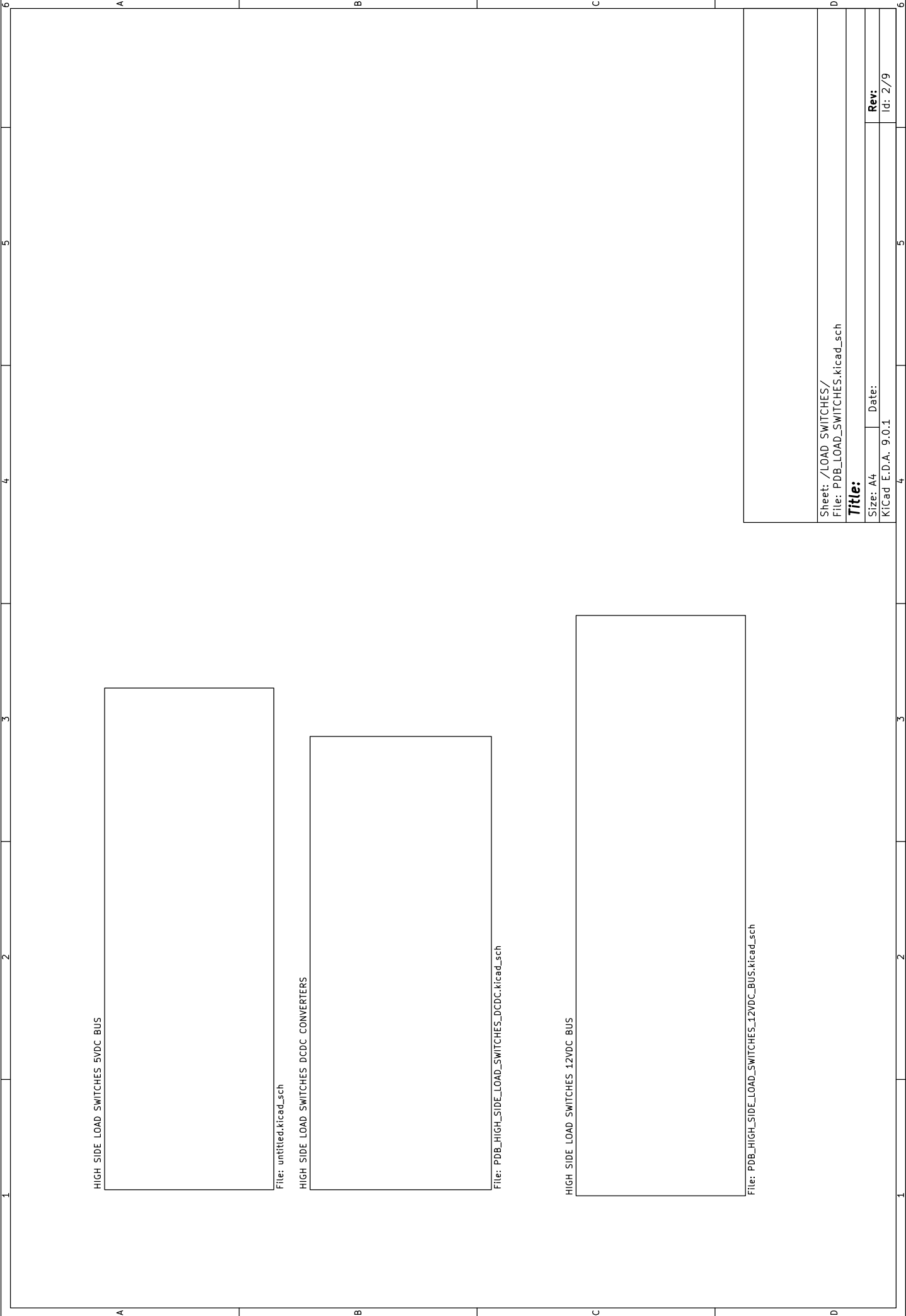
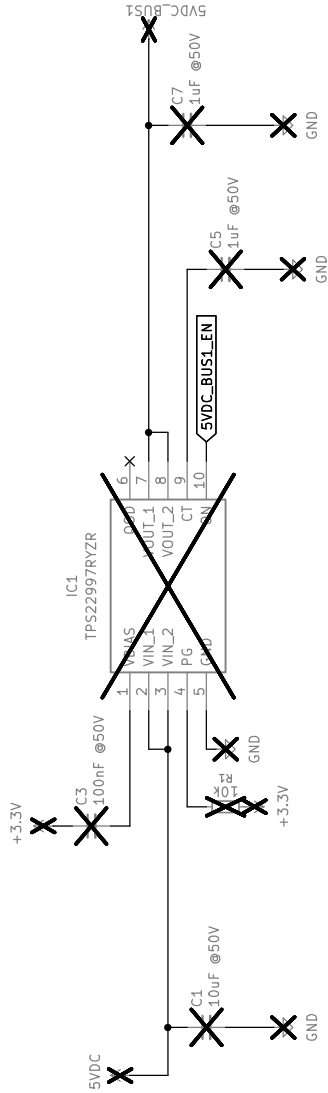


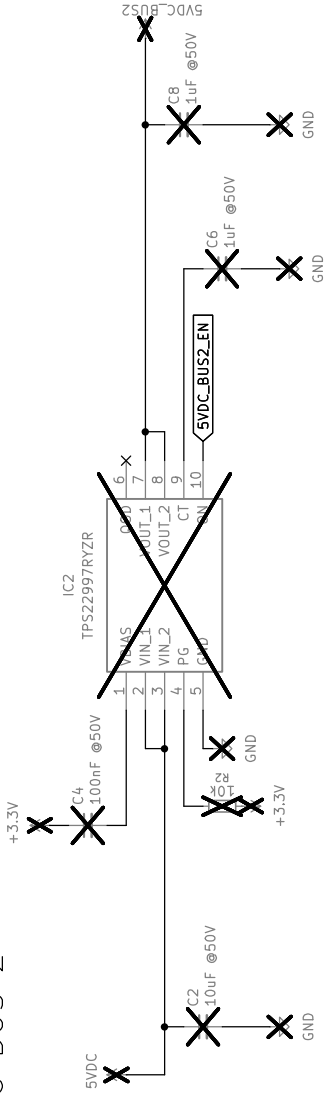
1	2	3	4	5	6			
A	LOAD SWITCHES		EXTERNAL CONNECTORS		A			
	File: PDB_LOAD_SWITCHES.kicad_sch		File: PDB_EXT_CONNECTORS.kicad_sch					
B	POWER ELECTRONICS		CURRENT SENSE		B			
	File: PDB_POWER_ELECTRONICS.kicad_sch		File: PDB_CURRENT_SENSE.kicad_sch					
C	LOW SIDE FETS					C		
	File: PDB_LOW_SIDE_FETS.kicad_sch							
D					D			
Sheet: /								
File: POWER DISTRIBUTIONB BOARD.kicad_sch								
Title:								
Size: A4			Date:		Rev:			
KiCad E.D.A. 9.0.1			Id: 1/9					
1	2	3	4	5	6			



5VDC BUS 1



5VDC BUS 2



Sheet: /LOAD SWITCHES/HIGH SIDE LOAD SWITCHES 5VDC BUS/
File: untitled.kicad_sch

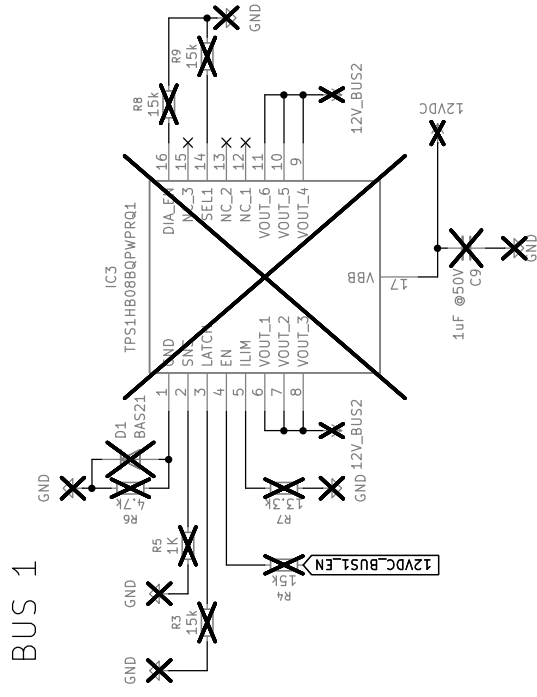
Title:

Size: A4 Date:

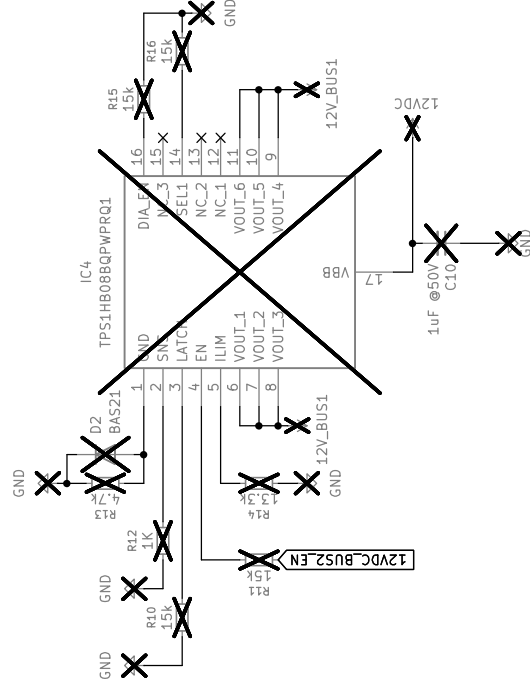
KiCad E.D.A. 9.0.1

Rev:

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BUS 1

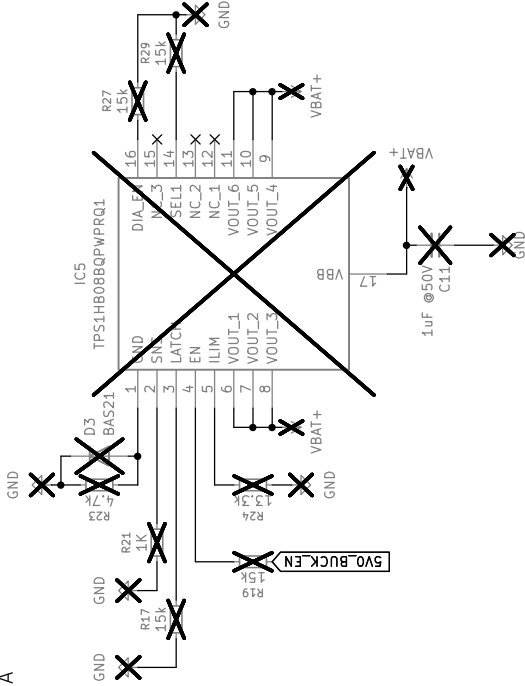


BUS 2

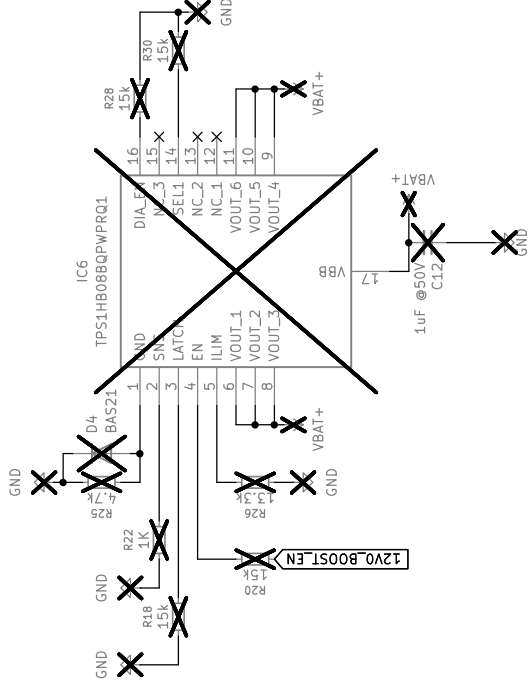
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File: PDB_High_SIDE_LOAD_SWITCHES_12VDC_BUS.kicad_sch	
Title:	
Size: A4	Date:
Rev:	
Id: 4/9	
KiCad E.D.A. 9.0.1	

5V0 BUCK CONVERTER

CURRENT_LIM=12A



12V0 BUCK CONVERTER



Sheet: /LOAD SWITCHES/HIGH SIDE LOAD SWITCHES DCDC CONVERTERS/
File: PDB_HIGH_SIDE_LOAD_SWITCHES_DCDC.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 9.0.1 Rev: Id: 5/9

CONNECTED TO MODE USING VOLTAGE DIVIDER

Voltage(V)

number required = +2



Required Capacitance: 66uF
Capacitance at DC bias of 12V: ~28.5
number required= ~3
USED: 22202 tdk



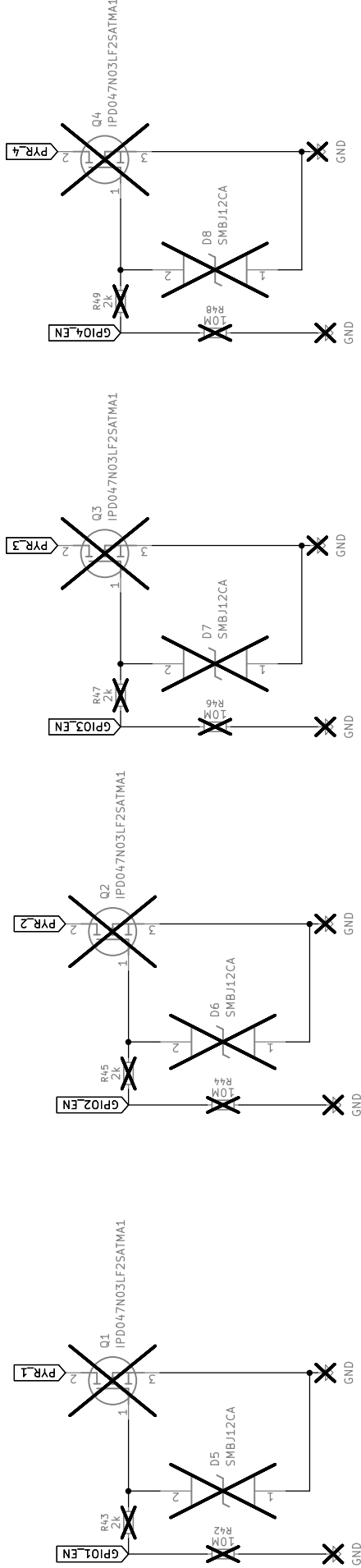
Sheet: /POWER_ELECTRONICS/
File: PDB_POWER_ELECTRONICS.kicad_sch

Size: A4	Date:
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Id: 7/9

ARMING MOSFET/LOAD SWITCHES

N CHANNEL MOSFETS ENHANCEMENT



Sheet: /LOW SIDE FETS/
File: PDB_LOW_SIDE_FETS.kicad_sch

Title:

Size: A4 Date:

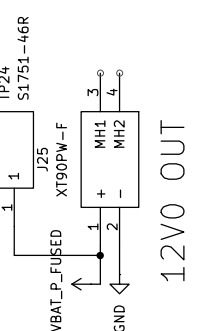
KiCad E.D.A. 9.0.1

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(PORT SIDE)

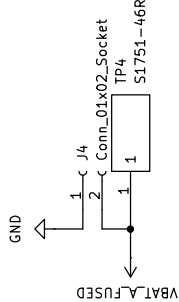
TIB PORT SIDE



12V0 OUT

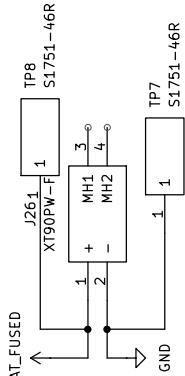
5V0 OUT

VBAT ACOUSTICS



(STARBOARD SIDE)

TIB STARBOARD SIDE

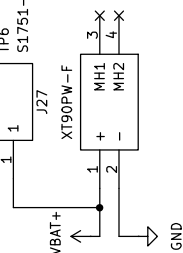


12V0 OUT

5V0 OUT

STERN

BAT1 CONN

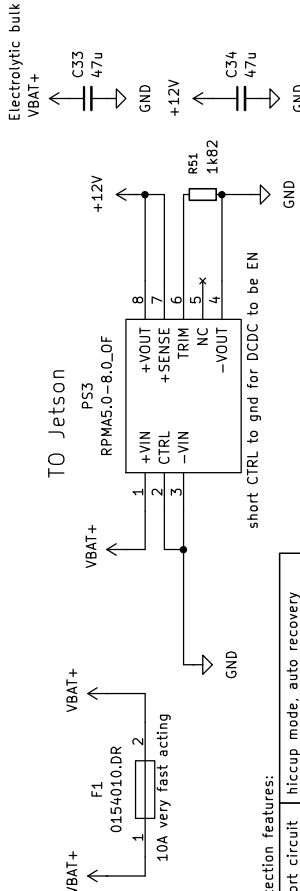


BAT1 CONN

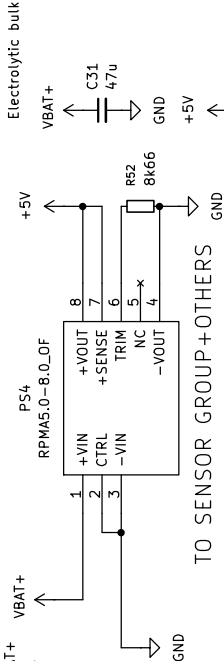
Mounting Holes: M3



BOW (ISOLATED GROUND)

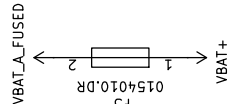
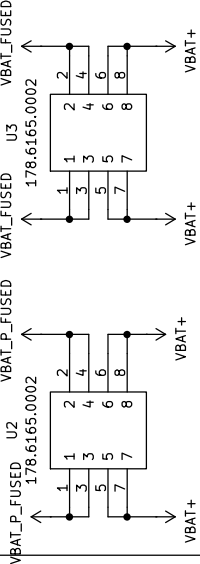


Protection features:	hiccup mode, auto recovery short circuit after fault removed
over-voltage	110%–150% of nominal Vout
Overload protection	150% typ (typ=6A)



TO SENSOR GROUP+OTHERS

In rush current calcs
over current protection: 14A @hiccup mode (fuse should blow at 10A)
Maximum Capacitive Load: 1200uF
Output current max= 6A, peak voltage=17.0V
Energy required =1/2*C*V^2=0.5*(1200e-06)*17^2= 1.734J



Sheet: /EXTERNAL CONNECTORS/
File: PDB_EXT_CONNECTORS.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.1

Rev:

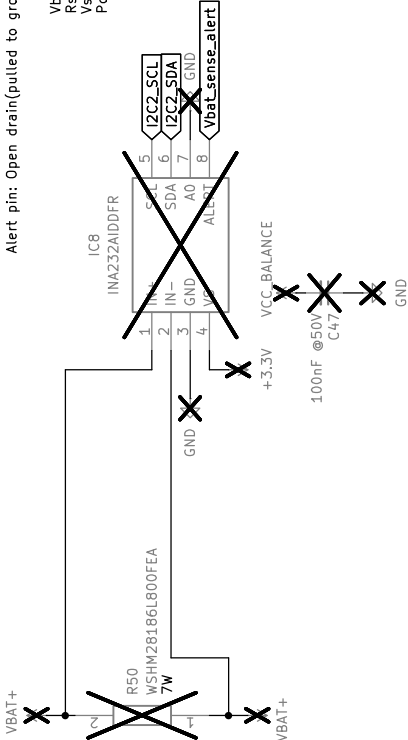
Id: 9/9

CURRENT_SENSE_VBAT

IDevice Address!
AO connected to VS:1000000
AO connected to GND:1000001

Alert pin: Open drain(pulled to ground) output, effective when limits exceeded or conversion ready

Vbat current sense:
Rshunt< Vsense/I max
Vsense= ±81.92 mV (ADCRANGE = 0). Imax=12[lim of all load switches], Rshunt=6.827mohm
Power= 6.8e-03*12 2=0.9792



Sheet: /CURRENT_SENSE/
File: PDB_CURRENT_SENSE.kicad_sch

Title:

Size: A4	Date:
KiCad E.D.A. 9.0.1	

Rev:

Id: 10/9